Quantum Mechanics II — Exercise Sheet 1

Energy and position in the box

Variant: Questions only

Exercise 1 (Reconstructing an energy distribution). In the box $(0, L)$ a unit vector lies in $\operatorname{span}\{u_1, u_2, u_3\}$. Its mean energy is $5E_1$ and its second energy moment is $31E_1^2$. Find all three energy probabilities and the energy variance. State what these data do and do not determine about the vector.

Exercise 2 (Position couples opposite parities). For $m, n \geq 1$ compute the matrix element $\langle u_m, Qu_n \rangle$. Treat $m = n$ separately. Use your result to decide which off-diagonal elements vanish. Here Q is multiplication by x in the box.

Exercise 3 (A sharp edge and finite energy). In the box let $\psi(x) = \sqrt{3/L}\,\mathbf{1}_{(0,L/3)}(x)$. (i) Find its sine coefficients and check their normalization by completeness. (ii) Decide whether the energy has a finite mean and whether $\psi \in D(H)$. Do not differentiate the discontinuous function as though it were smooth.

Exercise 4 (An energy filter). A normalized box state has coefficients $c_n = C/n^3$ for $n \geq 1$, with $C > 0$. Find C, its mean energy and the probability that a reading is at least $9E_1$. Give the normalized vector after that reading is retained. You may use $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$ and $\zeta(6) = \pi^6/945$.